Annals of Internal Medicine®

In the Clinic®

Alcohol Use

Health Benefits and Harms

Prevention and Screening

Diagnosis

Treatment

Practice Improvement

Tool Kit

Patient Information

Unhealthy alcohol use, the range of drinking that includes at-risk drinking and alcohol use disorder (1), is common and associated with a range of adverse health and social consequences. For example, data from the third National Epidemiologic Survey on Alcohol and Related Conditions (NESARC-III) indicate that 14% of adults have a current alcohol use disorder and 29% have had an alcohol use disorder over their lifetime, with rates highest among men and younger persons (2). As such, unhealthy alcohol use is the third leading cause of preventable death and costs the United States over $220 billion each year (3).

The CME quiz is available at www.annals.org/intheclinic.aspx. Complete the quiz to earn up to 1.5 CME credits.

Physician Writers
E. Jennifer Edelman, MD, MHS
David A. Fiellin, MD

doi: AITC201601050

CME Objective: To review current evidence for health benefits, harms, prevention, screening, diagnosis, treatment, and practice improvement of alcohol use.

Funding Source: American College of Physicians.

Disclosures: Dr. Edelman, ACP Contributing Author, has disclosed no conflicts of interest. Dr. Fiellin, ACP Contributing Author, reports personal fees from Pinney Associates outside the submitted work. Disclosures can also be viewed at www.acponline.org/authors/icmje/ConflictOfInterestForms.do?msNum=M15-2510. Dr. Edelman was funded as a Yale Drug Abuse, Addiction and HIV Research Scholars Program (K12DA033312-02) during the writing of this manuscript.

With the assistance of additional physician writers, the editors of ***Annals of Internal Medicine*** develop **In the Clinic** using **MKSAP** and other resources of the American College of Physicians.

In the Clinic does not necessarily represent official ACP clinical policy. For ACP clinical guidelines, please go to https://www.acponline.org/clinical_information/guidelines/.

© 2016 American College of Physicians

ACP American College of Physicians®
Leading Internal Medicine, Improving Lives

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

Although patients with unhealthy alcohol use commonly interface with the medical system, including through primary care, they often do not receive indicated care for this condition. The NESARC-III found that fewer than 4% of individuals with a current alcohol use disorder received treatment for this indication from a health care provider (2). To improve individual and public health by preventing and mitigating the harms of unhealthy alcohol use, clinicians providing care in diverse medical settings (i.e., primary care, inpatient) should be prepared to screen for unhealthy alcohol use and deliver effective treatments, including counseling and pharmacotherapy.

Health Benefits and Harms

Which health conditions have definite links to alcohol use?

Unhealthy alcohol use leads to myriad medical, psychiatric, and behavior-related complications and risks, which typically increase with higher levels of use (4, 5). Although methodological challenges and limitations in the literature make it difficult to accurately assess the effects of alcohol on common primary care conditions (6), systematic reviews have noted adverse effects on such disorders as hypertension (7), diabetes mellitus (8), osteoporosis (9), and depression (10) as well as an association with breast cancer (11).

Heavy episodic drinking can lead to acute alcohol poisoning, a medical emergency that results from high blood alcohol levels that suppress the central nervous system and can cause loss of consciousness, low blood pressure and body temperature, coma, respiratory depression, and death.

Hypertension, stroke, cardiomyopathy, cirrhosis, chronic pancreatitis, brain atrophy, hypogonadism with osteoporosis and sexual dysfunction, gastroesophageal reflux, esophagitis, peptic ulcers, pancreatitis, seizures, and arrhythmias are among the diseases associated with excess alcohol use. Even moderate alcohol use is carcinogenic and has been associated with increased risk for various types of cancer (including liver, mouth, throat, larynx, esophagus, breast, and colon) (11) and is an important reversible risk factor. Alcohol use is also independently associated with an increased incidence of certain chronic diseases, including diabetes mellitus (8) and HIV (12), and further complicates management given its negative effects on medication adherence (13).

Individuals with unhealthy alcohol use are often malnourished and are at risk for deficiencies in vitamin A, vitamin B complex, vitamin C, folic acid, carnitine, magnesium, selenium, zinc, essential fatty acids, and antioxidants.

A pregnant woman who drinks may harm the fetus. Complications including miscarriage, fetal alcohol syndrome, and more subtle neurocognitive consequences.

In addition to physical complications, unhealthy alcohol use causes mental health and social consequences. Alcohol use disorders are associated with depression (10), and heavy drinking episodes are associated with motor vehicle accidents, falls, drowning, burns, firearm injuries, unsafe sex, intimate partner violence, child maltreatment, homicide, and suicide.

Does alcohol use have positive health effects?

Although unhealthy alcohol use is associated with increased risk for cardiovascular disease, moderate alcohol use is protective. The mechanisms underlying this association are probably multifactorial

1. Saitz R. Clinical practice. Unhealthy alcohol use. N Engl J Med. 2005;352: 596-607. [PMID: 15703424]
2. Grant BF, Goldstein RB, Saha TD, Chou SP, Jung J, Zhang H, et al. Epidemiology of DSM-5 Alcohol Use Disorder: Results From the National Epidemiologic Survey on Alcohol and Related Conditions III. JAMA Psychiatry. 2015; 72:757-66. [PMID: 26039070] doi:10.1001/jamapsychiatry.2015.0584
3. National Institute on Alcohol Abuse and Alcoholism. Alcohol Facts and Statistics; 2015.Accessed at http://niaaa.nih.gov/alcohol-health/overview-alcohol-consumption/alcohol-facts-and-statistics on 9 September 2015.
4. Holt SR, Fiellin DA. Unhealthy alcohol use. Scientific American Medicine; 2014.
5. Berger D, Bradley KA. Primary Care Management of Alcohol Misuse. Med Clin North Am. 2015;99:989-1016. [PMID: 26320043] doi:10.1016/j.mcna.2015.05.004
6. Turner BJ, McLellan AT. Methodological challenges and limitations of research on alcohol consumption and effect on common clinical conditions: evidence from six systematic reviews. J Gen Intern Med. 2009;24: 1156-60. [PMID: 19672662] doi:10.1007/s11606-009-1072-z
7. McFadden CB, Brensinger CM, Berlin JA, Townsend RR. Systematic review of the effect of daily alcohol intake on blood pressure. Am J Hypertens. 2005;18: 276-86. [PMID: 15752957]
8. Howard AA, Arnsten JH, Gourevitch MN. Effect of alcohol consumption on diabetes mellitus: a systematic review. Ann Intern Med. 2004;140:211-9. [PMID: 14757619]
9. Berg KM, Kunins HV, Jackson JL, Nahvi S, Chaudhry A, Harris KA Jr, et al. Association between alcohol consumption and both osteoporotic fracture and bone density. Am J Med. 2008;121:406-18. [PMID: 18456037] doi:10.1016/j.amjmed.2007.12.012
10. Sullivan LE, Fiellin DA, O'Connor PG. The prevalence and impact of alcohol problems in major depression: a systematic review. Am J Med. 2005;118:330-41. [PMID: 15808128]

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

and may include development of favorable lipid profiles, inhibition of platelet activation, decreased fibrinogen levels, and anti-inflammatory effects (14).

A meta-analysis of 7 studies compared lifetime abstainers with persons with moderate alcohol use (without heavy episodic drinking) and found decreased risk for ischemic heart disease (pooled relative risk, 0.64 [95% CI, 0.53–0.71]) (14).

Although observational studies provide evidence that low levels of alcohol use protect against ischemic heart disease risk and similar associations have been seen for ischemic stroke (15), data from randomized, controlled trials are lacking (14); these studies may also be affected by unmeasured confounding (e.g., health care use). Given the overall effects of alcohol on health, these data should not translate into recommendations for initiation of low-level alcohol use for cardioprotective effects as indicated by the American Heart Association (16).

Which groups are at particularly high risk for adverse health outcomes from alcohol use?

The National Institute on Alcohol Abuse and Alcoholism (NIAAA) recognizes several groups as being particularly vulnerable to alcohol and its effects. This includes younger persons (i.e., college-aged and younger) and senior citizens (17). Senior citizens with a chronic medical condition (e.g., diabetes mellitus, congestive heart failure) who are receiving specific medications that may interact with alcohol (see the **Box**: Examples of Common Alcohol–Prescription Medication Interactions by Class Among Individuals With Current Alcohol Use) and those with polypharmacy, due to impaired metabolism and alcohol-medication interactions, are particularly vulnerable to the harms of alcohol use.

Notably, based on an analysis of data from the National Health and Nutrition Examination Survey, patients with current alcohol use are commonly prescribed medications that may interact with alcohol (adjusted prevalence, 42% [CI, 40%–43%]), and this is especially true among older adults (adjusted prevalence, 78% [CI, 76–80]) (18).

Women are more vulnerable to harms associated with alcohol because they have increased sensitivity. Any drinking during pregnancy can cause harm to the fetus. Minorities and underserved populations (e.g., those living in rural settings) often experience more alcohol-associated harms (e.g., cirrhosis-related deaths). In addition, regardless of age, patients with certain chronic medical conditions may be more sensitive to alcohol and its potential harms.

Examples of Common Alcohol–Prescription Medication Interactions by Class Among Individuals With Current Alcohol Use*

- Cardiovascular Agents
  - Angiotensin-converting enzyme inhibitors
  - β-blockers
  - Diuretics
- Central Nervous System Agents
  - Anticonvulsants
  - Anxiolytic/sedative/hypnotics
  - Opioids
  - Anticoagulants
- Metabolic Agents
  - Antidiabetics
  - Antihyperlipidemics
  - Antidepressants
  - Antihistamines
- Other
  - Antibiotics
  - Nonsteroidal anti-inflammatory agents

* From reference 18.

Health Benefits and Harms... Although moderate alcohol use may be cardioprotective, unhealthy alcohol use is associated with a range of adverse medical, psychiatric, and behavior-related outcomes. Special caution with alcohol use is appropriate among young adults, older adults, women, minorities and underserved populations, as well as those with chronic medical conditions and those who receive prescribed medications.

CLINICAL BOTTOM LINE

Prevention and Screening

When should clinicians screen for unhealthy alcohol use?

Given the negative consequences associated with unhealthy alcohol use, that it often goes unrecognized, and the availability of potentially effective treatments, the NIAAA recom-

11. Scoccianti C, Cecchini M, Anderson AS, Berrino F, Boutron-Ruault MC, Espina C, et al. European Code against Cancer 4th Edition: Alcohol drinking and cancer.Cancer Epidemiol. 2015 Jun 24 [Epub ahead of print]. [PMID: 26115567] doi: 10.1016/j.canep.2015.01.007.
12. Baliunas D, Rehm J, Irving H, Shuper P. Alcohol consumption and risk of incident human immunodeficiency virus infection: a meta-analysis. Int J Public Health. 2010;55:159-66. [PMID: 19949966] doi:10.1007/s00038-009-0095-x
13. Grodensky CA, Golin CE, Ochtera RD, Turner BJ. Systematic review: effect of alcohol intake on adherence to outpatient medication regimens for chronic diseases. J Stud Alcohol Drugs. 2012;73:899-910. [PMID: 23036207]
14. Roerecke M, Rehm J. Alcohol consumption, drinking patterns, and ischemic heart disease: a narrative review of meta-analyses and a systematic review and meta-analysis of the impact of heavy drinking occasions on risk for moderate drinkers. BMC Med. 2014;12:182. [PMID: 25567363] doi:10.1186/s12916-014-0182-6

 © 2016 American College of Physicians

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

National Institute on Alcohol Abuse and Alcoholism–Recommended Screening Opportunities

Routine examination

Before prescribing a medication with potential interactions with alcohol

In the emergency department or urgent care center

When seeing a patient who:

- Is pregnant or planning conception
- Has risk factors for unhealthy alcohol use (e.g., smokes tobacco, young adults)
- Has potentially alcohol-related health problems (e.g., arrhythmia, cirrhosis, trauma)
- Has a chronic condition resistant to usual treatment (e.g., pain, depression, diabetes mellitus, hypertension)

mends routine screening, with a focus on particular clinical encounters and patients (see the **Box**: National Institute on Alcohol Abuse and Alcoholism–Recommended Screening Opportunities) (19). Similarly, the U.S. Preventive Services Task Force (USPSTF) and the Centers for Disease Control and Prevention (CDC) recommend routine screening of adults for unhealthy alcohol use and provision of brief counseling for those with at-risk drinking (20, 21).

What self-report–based methods are effective for screening for unhealthy alcohol use in clinical settings?

The NIAAA recommends first asking patients, "Do you sometimes drink beer, wine, or other alcoholic beverages?" and then, for men who respond affirmatively, asking, "How many times in the past year have you had 5 or more drinks in a day?" and for women who respond affirmatively, asking, "How many times in the past year have you had 4 or more drinks in a day?" (19). Alternatively, patients may be requested to complete a standardized instrument, such as the AUDIT (Alcohol Use Disorders Identification Test) or AUDIT-C (**Appendix Figure**, available at www.annals.org), an approach that has been widely implemented in primary care settings (19, 22). Although the USPSTF and CDC recommend a similar approach (20, 21), it is important to recognize that the operating characteristics of the AUDIT and AUDIT-C vary based on clinical setting and cut-offs used (19, 23). Further, implementation of these instruments is optimized in clinical settings when the instrument is used as intended (i.e., verbatim language) with standardized clinical reminders and processes in place (24). The SISQ [Single-Item Screening Question]-Alcohol instrument from the NIAAA can be used to streamline the screening process—it includes only 1 question: "How many times in the past year have you had *x* or more drinks in a day?" (where *x* is 5 for men and 4 for women).

A recent study compared computer-administered SISQ-Alcohol instrument with research assistant–administered standardized measures (e.g., timeline follow-back, Short-Inventory of Problems and MINI-Plus) to assess alcohol use and alcohol-related problems. Based on the evaluation of 459 participants from 2 urban primary care clinics, the SISQ-Alcohol instrument had a sensitivity of 73.3% (CI, 65.3%–80.3%), a specificity of 84.7% (CI, 80.2%–88.5%), and an area under the curve (AUC) of 0.79 (CI, 0.75–0.83) for detecting unhealthy alcohol use and sensitivity of 86.7% (CI, 75.4–94.1), specificity of 74.2% (CI, 69.6–78.4), and AUC of 0.80 (CI, 0.76–0.85), for alcohol use disorder (25).

SISQ-Alcohol, thus, holds promise as a valid approach for detecting unhealthy alcohol use in primary care settings.

Patients with a negative screen, should be counseled regarding guidelines for continued low-risk drinking or abstinence as appropriate (e.g., cirrhosis, history of alcohol use disorder). Patients with a positive screen should have further assessment of the frequency (i.e., "On average, how many days a week do you have an alcoholic drink?") and quantity (i.e., "On a typical drinking day, how many drinks to do you have?") of alcohol use as well as the number of heavy drinking days, followed by careful assessment for an alcohol use disorder (19) and alcohol-related consequences.

Although other instruments, such as the CAGE (Cut down, Annoyed, Guilty, Eye-opener) questionnaire and the MAST (Michigan Alcohol Screening Test) have been validated and widely implemented, they are more appropriate for screening for lifetime alcohol use disorders than for

15. Patra J, Taylor B, Irving H, Roerecke M, Baliunas D, Mohapatra S, et al. Alcohol consumption and the risk of morbidity and mortality for different stroke types–a systematic review and meta-analysis. BMC Public Health. 2010;10:258. [PMID: 20482788] doi:10.1186/1471-2458-10-258
16. American Heart Association. Alcohol and Heart Health; 2015. Accesssed at www.heart.org/HEARTORG/Conditions/More/MyHeartandStrokeNews/Alcohol-and-Heart-Disease_UCM_305173_Article.jsp on 15 September 2015.
17. National Institute on Alcohol Abuse and Alcoholism.Special Populations & Co-occurring Disorders. Accessed at www.niaaa.nih.gov/alcohol-health/special-populations-co-occurring-disorders on 15 September 2015.
18. Breslow RA, Dong C, White A. Prevalence of alcohol-interactive prescription medication use among current drinkers: United States, 1999 to 2010. Alcohol Clin Exp Res. 2015;39:371-9. [PMID: 25597432] doi:10.1111/acer.12633

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

lower levels of problem drinking or binge drinking and less able to distinguish current from prior alcohol use disorders.

Patients should be screened in a nonjudgmental manner that optimizes comfort and helps them understand the relevance of alcohol use to their health and occurs in the context of inquiry about other health-related behaviors (e.g., exercise, nutrition, tobacco use) (4, 5). As patterns of alcohol use change over time, repeated screening is important but may be less necessary if results are negative on multiple consecutive occasions, particularly among women and older adults (26).

Prevention and Screening... Standardized processes involving validated instruments and approaches should be incorporated into practices to facilitate routine screening of all patients for unhealthy alcohol use. Patients with negative screening results should be counseled on maintaining lower-risk alcohol use or abstinence as appropriate. Those with positive results should be evaluated for alcohol use disorders and alcohol-related consequences and then provided appropriate treatment.

CLINICAL BOTTOM LINE

Diagnosis

How should clinicians distinguish between lower-risk alcohol consumption, at-risk drinking, and alcohol use disorders?

Although different definitions have been applied (19, 27), according to the NIAAA alcohol use can be broadly classified into 1 of 3 categories. These categories are essential to determining risk for specific conditions and guiding appropriate preventative and treatment strategies for patients. Most common is the so-called *moderate* or *lower-risk alcohol use without health consequences*, which is associated with lowest risk for alcohol-related consequences. According to the NIAAA, for men up to age 65 years lower-risk alcohol use is no more than 4 drinks on any single day and no more than 14 drinks per week. For women, who tend to have lower volumes of distribution, and men older than 65 years, lower-risk alcohol use is no more than 3 drinks on any single day and no more than 7 drinks per week (19).

Hazardous or at-risk drinking is the pattern of alcohol use associated with increased risk for alcohol-related consequences and occurs when the thresholds for lower-risk alcohol use are exceeded (19) or when drinking in lower amounts increases risk (e.g., pregnancy). Heavy drinking is defined as more than 4 drinks on any day for men and more than 3 drinks on any day for women (28).

As defined by the World Health Organization and included in the International Classification of Diseases and Related Health Problems, 10th revision (ICD-10), *harmful alcohol use* is a pattern of drinking that causes health consequences (27).

*Alcohol use disorder, as defined by the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), is present if an individual meets at least 2 of the 11 criteria (see the **Box**: DSM-5 Criteria for Alcohol Use Disorder) (29). Determining severity is important because patients with a moderate or severe alcohol use disorder (meeting 3 or more criteria) may benefit from more intensive treatment. This reflects a change from the DSM-IV, which included separate diagnoses for alcohol abuse and alcohol dependence.*

Unhealthy alcohol use refers to persons with hazardous/at-risk drinking, harmful drinking, or alcohol use disorder (1).

19. National Institute on Alcohol Abuse and Alcoholism. Helping Patients Who Drink Too Much: A Clinician's Guide; 2005. Accessed at pubs.niaaa.nih.gov/.../CliniciansGuide2005/guide.pdf on 13 November 2015.
20. Jonas DE, Garbutt JC, Amick HR, Brown JM, Brownley KA, et al. Behavioral counseling after screening for alcohol misuse in primary care: a systematic review and meta-analysis for the U.S. Preventive Services Task Force. Ann Intern Med. 2012;157:645-54. [PMID: 23007881]
21. Centers for Disease Control and Prevention. Planning and Implementing Screening and Brief Intervention for Risky Alcohol Use: A Step-by-Step Guide for Primary Care Practices. Atlanta, Ga: Centers for Disease Control and Prevention, National Center on Birth Defects and Developmental Disabilities; 2014.
22. Bradley KA, Williams EC, Achtmeyer CE, Volpp B, Collins BJ, Kivlahan DR. Implementation of evidence-based alcohol screening in the Veterans Health Administration. Am J Manag Care. 2006; 12:597-606. [PMID: 17026414]
23. Reinert DF, Allen JP. The alcohol use disorders identification test: an update of research findings. Alcohol Clin Exp Res. 2007;31:185-99. [PMID: 17250609]
24. Williams EC, Achtmeyer CE, Thomas RM, Grossbard JR, Lapham GT, Chavez LJ, et al. Factors Underlying Quality Problems with Alcohol Screening Prompted by a Clinical Reminder in Primary Care: A Multisite Qualitative Study. J Gen Intern Med. 2015; 30:1125-32. [PMID: 25731916] doi:10.1007/s11606-015-3248-z
25. McNeely J, Cleland CM, Strauss SM, Palamar JJ, Rotrosen J, Saitz R. Validation of Self-Administered Single-Item Screening Questions (SISQs) for Unhealthy Alcohol and Drug Use in Primary Care Patients. J Gen Intern Med. 2015;30: 1757-64. [PMID: 25986138] doi:10.1007/s11606-015-3391-6

 © 2016 American College of Physicians

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

DSM-5 Criteria for Alcohol Use Disorder*

- Alcohol taken in larger amounts or for longer than intended
- Persistent desire or unsuccessful efforts to cut down or control alcohol use
- Great deal of time spent obtaining, using, or recovering from alcohol use
- Craving or strong desire to use alcohol
- Failure to fulfill major obligations due to alcohol use
- Continued use despite problems caused or exacerbated by alcohol use
- Important activities given up or reduced because of alcohol use
- Recurrent alcohol use in physically hazardous situations
- Continued use despite knowledge of physical or psychological problems that are caused or exacerbated by alcohol
- Tolerance
- Withdrawal

DSM-5 = Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition.

* Mild = 2–4 symptoms; moderate = 4–5 symptoms; severe = 6 or more symptoms.

Screening and assessment are the methods by which a patient's level of alcohol use may be appropriately categorized. Patients are classified as having an alcohol use disorder if they have met at least 2 criteria for an alcohol use disorder in the past 12 months. Otherwise, they are considered to have at-risk drinking. *Classifying alcohol use accurately is important because it directly guides treatment options.*

As appropriate, treatment history, family history of substance use disorders and mental illness, and alcohol-related consequences should be assessed.

What is the role of the physical examination and laboratory testing in the evaluation of patients with unhealthy alcohol use?

In conjunction with self-reported data collected in the history, the physical examination and laboratory testing may be helpful in identifying and evaluating patients with unhealthy alcohol use. For example, patients with evidence of worsening hypertension or tachycardia may be manifesting withdrawal. Signs and symptoms consistent with liver, cardiac, or neurocognitive disease, may signal consequences of alcohol use (see the **Box**: Findings That May Indicate Unhealthy Alcohol Use).

To date, there are no widely available biomarkers that reliably measure alcohol use and predict the impact of alcohol use on health. Increased mean corpuscular volume of red blood cells, elevated γ-glutamyltransferase, and increased aspartate–to–alanine aminotransferase ratio may signal unhealthy alcohol use, but sensitivity and specificity vary (30). Markers related to ethanol metabolism, such as phosphatidyl ethanol and carbohydrate-deficient transferrin, are under investigation (31).

Which other conditions should clinicians watch for in patients with unhealthy alcohol use?

Substance use disorders, especially tobacco use disorders, and mental illness commonly co-occur with unhealthy alcohol use.

An analysis of NESARC-III data (n = 36 309) showed that patients with any alcohol use disorder had an 80% increased risk for a major depressive disorder (adjusted odds ratio [OR], 1.8 [CI, 1.6–2.0]) (2). This risk was greatest among those with a severe alcohol use disorder (adjusted OR, 2.9 [CI, 2.5–3.5]). The odds of mental illness increased with severity of alcohol use disorder across multiple diagnoses, including other mood disorders, anxiety disorders, posttraumatic stress disorder, and personality disorders.

Similarly, chronic pain and unhealthy alcohol use often co-occur. Patients should be carefully evaluated for these underlying conditions because their presence can affect treatment decisions and response. As sexual risk behaviors are common in the setting of unhealthy alcohol use, patients should be evaluated accordingly.

Diagnosis... Diagnosis relies on a comprehensive evaluation, including history, physical examination, and supporting laboratory data, as well as patient self-reported information. Although no single laboratory test reliably measures alcohol use and its effects, various markers can be used to detect alcohol use and measure its impact on health. Given the increased prevalence of comorbid conditions among patients with unhealthy alcohol use (e.g., substance use, mental illness, and chronic pain), efforts to screen for and address these conditions in addition to sexual risk behavior should be part of routine care.

CLINICAL BOTTOM LINE

26. Lapham GT, Rubinsky AD, Heagerty PJ, Williams EC, Hawkins EJ, Maynard C, et al. Annual rescreening for alcohol misuse: diminishing returns for some patient subgroups. Med Care. 2013;51:914-21. [PMID: 23969582] doi:10.1097/MLR.0b013e3182a3e549
27. Babor TF, Higgins-Biddle JC. Brief Intervention for Hazardous and Harmful Drinking. A Manual for Use in Primary Care. Geneva: World Health Organization; 2001.
28. National Institute on Alcohol Abuse and Alcoholism. What's "At-Risk" or "Heavy" Drinking? Accessed at http://rethinkingdrinking.niaaa.nih.gov/IsYourDrinkingPatternRisky/WhatsAtRiskOrHeavyDrinking.asp on 11 Nov 2015.

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

Treatment approaches are guided by the patient's level of alcohol use and diagnostic category. Brief interventions advising less or no use are appropriate for those with at-risk alcohol use; for those with disorders, advice should include engaging in more intensive treatment. More comprehensive treatment approaches, including management of withdrawal symptoms and behavioral interventions with pharmacotherapy, may be indicated for persons with an alcohol use disorder.

What language is appropriate during treatment of patients with unhealthy alcohol use?

Regardless of the level of alcohol use, caution should be used to avoid imprecise and stigmatizing language. For instance, "problem drinking" has various definitions, making its use unsuitable. In contrast, a term such as alcohol "abuse" refers to a specific DSM diagnosis but is often used in a generic sense to refer to unhealthy alcohol use or alcohol use disorder. To avoid stigma, help focus on the medical aspects of unhealthy alcohol use, and facilitate appropriate treatment, the following practices have been advocated: use *people-first* language (i.e., *a person with an alcohol use disorder*), focus on the medical aspects of the condition and its treatment, and avoid use of slang and idioms (32). Thus, such terms as *alcoholic* and *alcohol abuser* should be avoided (32, 33). When considering and discussing unhealthy alcohol use, it can be helpful to make the comparison to another chronic medical condition, such as diabetes mellitus, where the cause is also based on a combination of genetic and behavioral factors and treatment requires a comprehensive approach (32, 34).

What should clinicians do if they identify patients with hazardous or at-risk alcohol use?

The goal for patients with hazardous or at-risk alcohol use is to decrease alcohol consumption to moderate levels to prevent alcohol-related harms. Brief interventions, 5–20 minutes in duration that are effective when delivered by physicians and nonphysicians alike, are feasible and effective in primary care settings (35). These interventions are designed to elicit the patient's perception of his or her alcohol use and its associated risks and should include the following: clear advice with a specific recommendation regarding alcohol use; personalized and normative feedback regarding the effects of alcohol on the patient's health and his or her alcohol use relative to norms; and empathy with acknowledgment of the patient's ability and responsibility in making a change. Patients who express an interest in change should be provided a menu of options; situations that are likely to trigger excessive alcohol use should be discussed; and a drinking agreement and follow-up should be arranged (1, 19). Individuals who are not ready to change their alcohol use should be given specific advice about recommended alcohol use; asked about their reasons for both drinking and avoiding alcohol (i.e., pros and cons of alcohol use); and asked what it would take to motivate change (i.e., increase readiness to change on 1–10 scale), with a plan for follow-up evaluation (1).

*A systematic review evaluated the efficacy of behavioral counseling interventions (most of which were brief) done in primary care targeting persons with at-risk drinking (*n *= 23). The authors found that compared with controls,*

Treatment

Findings That May Indicate Unhealthy Alcohol Use

Hypertension
Jaundice
Spider angiomata
Cardiomyopathy
Atrial fibrillation
Gynecomastia
Hepatosplenomegaly
Ascites
Testicular atrophy
Palmar erythema, plethoric facies
Peripheral neuropathy
Cognitive abnormalities

29. American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition. Washington, DC: American Psychiatric Press; 2013.
30. Hannuksela ML, Liisanantti MK, Nissinen AE, Savolainen MJ. Biochemical markers of alcoholism. Clin Chem Lab Med. 2007;45:953-61. [PMID: 17579567]
31. Wurst FM, Thon N, Yegles M, Schrück A, Preuss UW, Weinmann W. Ethanol Metabolites: Their Role in the Assessment of Alcohol Intake. Alcohol Clin Exp Res. 2015;39:2060-72. [PMID: 26344403] doi:10.1111/acer.12851. 2015.
32. Broyles LM, Binswanger IA, Jenkins JA, Finnell DS, Faseru B, Cavaiola A, et al. Confronting inadvertent stigma and pejorative language in addiction scholarship: a recognition and response [Editorial]. Subst Abus. 2014;35:217-21. [PMID: 24911031] doi:10.1080/08897077.2014.930372
33. Kelly JF, Wakeman SE, Saitz R. Stop talking 'dirty': clinicians, language, and quality of care for the leading cause of preventable death in the United States [Editorial]. Am J Med. 2015;128:8-9. [PMID: 25193273] doi:10.1016/j.amjmed.2014.07.043
34. Samet JH, Fiellin DA. Opioid substitution therapy-time to replace the term [Letter]. Lancet. 2015;385:1508-9. [PMID: 25933278] doi:10.1016/S0140-6736(15)60750-4

 © 2016 American College of Physicians

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

Signs and Symptoms of Alcohol Withdrawal*

Minor symptoms: Diaphoresis, nystagmus, tachycardia, hyperreflexia, hypertension, nausea or vomiting, low-grade fever, diarrhea, mild agitation

Hallucinations (auditory, visual, tactile): May occur while intoxicated; sensorium otherwise clear unless progression to delirium tremens

Withdrawal seizures: Grand mal, peak occurrence 12–48 hours after last drink; cluster may occur over 3–6 hours, although last seizure occurs more than 6 hours after first in 15% of cases

Delirium tremens: Agitated confusional state with tremulousness, hallucinations, and striking autonomic overactivity; fever in 82% of cases, often associated with comorbid illness

* From reference 39.

those receiving the counseling intervention decreased alcohol consumption by 3.6 drinks per week from baseline; 12% fewer adults reported heavy drinking episodes, and 11% more adults reported drinking less than the recommended limits over 12 months (35).

Notably, brief, multicontact interventions were most effective (19). Electronic brief interventions have been studied but to date have not been shown to be routinely effective (36).

How should care of patients with an alcohol use disorder be prioritized?

Patients with alcohol use disorder seen in primary care settings may present with a wide range of treatment needs, including management of withdrawal symptoms as well as prevention of relapse. Care should be prioritized to promote patient safety and stabilization first. Although reduction in alcohol consumption is the initial goal for patients with alcohol use disorder, few are able to maintain controlled drinking and so abstinence is typically the goal of therapy.

How should alcohol withdrawal be addressed in the primary care setting?

Among patients with an alcohol use disorder, withdrawal may manifest with a range of signs and symptoms, including abnormal vital signs (i.e., elevated heart rate, blood pressure, temperature), autonomic hyperactivity, gastrointestinal symptoms, and central nervous system effects (see the **Box**: Signs and Symptoms of Alcohol Withdrawal) (4, 37). Some patients can safely be managed in the outpatient setting with close follow-up, whereas patients at greater risk for harm (e.g., severe withdrawal with delirium tremens, seizures, concomitant drug use, elderly persons) or those who are unlikely to follow up should be referred to an inpatient setting (see the **Box**: Indications for Referral for Inpatient Detoxification) (38).

Standardized instruments may be used to assess the degree of withdrawal and guide treatment. The Clinical Institute Withdrawal Assessment for Alcohol, Revised (CIWA-Ar) (39), which includes 10 items that are summed to create a score, is most commonly used. A score less than 8 indicates mild withdrawal, a score of 8–15 indicates moderate withdrawal, and a score greater than 15 indicates severe withdrawal. These scores can be used to guide changes in clinical status and medication management.

Multiple dosing strategies and medication regimens have been previously evaluated for the treatment of alcohol withdrawal symptoms and seizure prevention (4, 40). The use of benzodiazepines dosed according to symptoms (i.e., symptom-triggered) is the safest and most effective strategy and is preferred over such alternatives as anticonvulsants (41–43). Typically, long-acting benzodiazepines (e.g., chlordiazepoxide and diazepam) are preferred; however, short-acting benzodiazepines should be considered in older adults and those with liver disease. Additional medications may be used as needed for symptom management, such as α_2-agonists and β-blockers for autonomic hyperactivity and neuroleptics for hallucinations (44). Other agents, such as topiramate, gabapentin, and baclofen, remain under investigation for use in managing alcohol withdrawal (44).

Because withdrawal can begin as early as 5 to 8 hours and as late as 72 hours after the last drink (40), patients should be monitored closely during this period

35. Jonas DE, Garbutt JC, Amick HR, Brown JM, Brownley KA, Council CL, et al. Behavioral counseling after screening for alcohol misuse in primary care: a systematic review and meta-analysis for the U.S. Preventive Services Task Force. Ann Intern Med. 2012;157: 645-54. [PMID: 23007881]
36. Dedert EA, McDuffie JR, Stein R, McNiel JM, Kosinski AS, Freiermuth CE, et al. Electronic Interventions for Alcohol Misuse and Alcohol Use Disorders: A Systematic Review. Ann Intern Med. 2015;163:205-14. [PMID: 26237752] doi:10.7326/M15-0285
37. Goodson CM, Clark BJ, Douglas IS. Predictors of severe alcohol withdrawal syndrome: a systematic review and meta-analysis. Alcohol Clin Exp Res. 2014;38: 2664-77. [PMID: 25346507] doi:10.1111/acer.12529
38. Fiellin DA, Reid MC, O'Connor PG. Outpatient management of patients with alcohol problems. Ann Intern Med. 2000; 133:815-27. [PMID: 11085845]

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

and then bridged to treatment for relapse prevention.

What is the role of psychotherapeutic interventions for patients with an alcohol use disorder?

For patients with an alcohol use disorder, as well as those with at-risk alcohol use who are unable to meet treatment goals, psychotherapeutic interventions are a mainstay of treatment. Cognitive behavioral therapy, motivational enhancement therapy, and 12-step facilitation are common treatments (38) and have similar efficacy (45).

Cognitive behavioral therapy has 2 main components. First, through functional analysis, patients are guided to identify the thoughts, feelings, and circumstances that occur before and after alcohol use. The goal is to help them understand why they consume alcohol, identify coping difficulties, and determine triggers for relapse. Then, through skills-based training, patients develop new behaviors and techniques for coping with these triggers.

Motivational enhancement therapy, grounded in stages of change theory, helps motivate patients to change their alcohol use.

Twelve-step facilitation involves a manual-driven approach to facilitate recovery with the underlying premise that alcohol use disorders are secondary to a medical and spiritual disease. Twelve-step facilitation is a formal process of facilitating a patient's engagement in Alcoholics Anonymous (AA), as opposed to a simple referral.

Other psychotherapeutic interventions, such as community reinforcement and behavioral couples therapy, are also supported by the literature (46).

Mutual-help groups, including AA, represent an additional treatment option for patients with alcohol use disorder. AA is the most well-known of these groups focused on alcohol, yet other groups exist. Data on the effectiveness of AA conflict (46). *However, a recent analysis, using instrumental variables models to analyze data from 6 randomized clinical trials found that for most individuals, AA attendance was associated with increasing days of abstinence at 3- and 15-month follow-up (47).*

Because AA is free, widely accessible, and only requires a desire to stop drinking, patients should be routinely referred and encouraged to attend at least 1 meeting. Strategies that may improve engagement include increased provider familiarity with local groups, stocking AA literature, and following up on a patient's experience after referral (48).

When should clinicians consider pharmacotherapy for relapse prevention for patients with an alcohol use disorder?

Pharmacotherapy for relapse prevention should be considered for all patients with an alcohol use disorder (49), particularly those with a moderate to severe disorder (**Table**). There are 3 U.S. Food and Drug Administration (FDA)–approved medications that should be considered for treatment in patients with alcohol use disorders: disulfiram, acamprosate, and naltrexone. Outcomes with these medications are typically best in conjunction with a minimum of brief psychosocial counseling (50). For acamprosate and naltrexone, a brief period of abstinence before initiation is not mandatory but is associated with improved outcomes.

Indications for Referral for Inpatient Detoxification

- Reasons for immediate referral for inpatient detoxification
  - Moderate to severe withdrawal
  - History of seizures or delirium tremens
  - Unable to adhere to daily follow-up
  - Comorbid psychiatric or medical complications requiring hospitalization
  - Unable to take oral medication
  - Unsuccessful outpatient detoxification
  - Pregnancy
- Reasons to strongly consider inpatient detoxification
  - Coexisting benzodiazepine use
  - High risk for severe alcohol withdrawal, including older age, heavy drinking for an extended period, consuming more than 100 g ethanol daily, random blood alcohol level greater than 200 mg/dL, signs and symptoms of alcohol withdrawal when not drinking

39. Sullivan JT, Sykora K, Schneiderman J, Naranjo CA, Sellers EM. Assessment of alcohol withdrawal: the revised clinical institute withdrawal assessment for alcohol scale (CIWA-Ar). Br J Addict. 1989;84:1353-7. [PMID: 2597811]
40. Blondell RD. Ambulatory detoxification of patients with alcohol dependence. Am Fam Physician. 2005;71:495-502. [PMID: 15712624]
41. Amato L, Minozzi S, Vecchi S, Davoli M. Benzodiazepines for alcohol withdrawal. Cochrane Database Syst Rev. 2010: CD005063. [PMID: 20238336] doi:10.1002/14651858.CD005063.pub3
42. Minozzi S, Amato L, Vecchi S, Davoli M. Anticonvulsants for alcohol withdrawal. Cochrane Database Syst Rev. 2010: CD005064. [PMID: 20238337] doi:10.1002/14651858.CD005064.pub3

 © 2016 American College of Physicians

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

Table. **Pharmacotherapy for Patients With an Alcohol Use Disorder**

*Medication (Typical Dosage)**	*Indication*	*Mechanism*	*Side Effects*	*Notes*
Benzodiazepines (Symptom-triggered: chlordiazepoxide, 50–100 mg; diazepam, 10–20 mg; or lorazepam, 2–4 mg every 1–2 hours until symptoms subside Fixed-dose: chlordiazepoxide, 50 mg; diazepam, 10 mg; or lorazepam, 2 mg every 6 hours on day 1, then one half dose every 6 hours on days 2 and 3)	Treatment or prophylaxis for alcohol withdrawal syndrome	Enhances GABA inhibition of neuronal excitability	Oversedation, paradoxic hyperactivity, depression. Potential for addiction.	Caution in the presence of respiratory or hepatic impairment
Naltrexone (oral 50–100 mg daily or injectable 380 mg monthly)	Relapse prevention	Opioid antagonist that may reduce the subjective reward associated with alcohol use	Nausea, indigestion, headache, fatigue. Depressive symptoms. Rarely medication-associated hepatitis. Potential for precipitated opioid withdrawal if opioids present.	Contraindicated in the presence of opioid use. Avoid if decompensated cirrhosis; use with caution with hepatitis, compensated cirrhosis.
Acamprosate (666 mg 3 times daily)	Relapse prevention	May antagonize glutamate-mediated neuronal hyperexcitability and reduce prolonged (but not acute) withdrawal symptoms	Diarrhea, nausea/vomiting, myalgia, rash, dizziness, palpitations. Rarely associated with renal impairment.	Reduced dosage with renal insufficiency. May be used with naltrexone. Medication adherence may be challenging.
Disulfiram (250–500 mg daily)	Drinking and relapse prevention	Aldehyde dehydrogenase inhibition results in acetaldehyde accumulation with alcohol use, leading to unpleasant symptoms (i.e., alcohol-disulfiram reaction)	Drowsiness, rash. Rarely medication-associated severe hepatotoxicity, optic neuritis, peripheral neuropathy.	Potential for many medication–medication interactions. Patient must be abstinent at least 12 hours before administration. Avoid in patients with hepatic impairment or cardiovascular disease. Most appropriate for patients with strong motivation to be abstinent and with support to promote medication adherence.

GABA = γ-aminobutyric acid.
** Naltrexone, disulfiram, and acamprosate are U.S. Food and Drug Administration pregnancy category C (animal studies indicate potential fetal risk or have not been conducted and no or insufficient human studies have been done; drugs in this category should be used in pregnant or lactating women only when potential benefits justify potential risk to the fetus or infant). Benzodiazepines are category X (contraindicated in pregnancy) or D (positive evidence of risk).*

Disulfiram, approved by the FDA in 1951, was the first approved medication for alcohol use disorder. It irreversibly inhibits the activity of acetaldehyde dehydrogenase, leading to buildup of acetaldehyde after ethanol ingestion. This medication is appropriate when the initial goal is abstinence because the adverse side effects resulting from alcohol use, including nausea, flushing, and palpitations, act as a deterrent. *Although models with selected patient populations have demonstrated efficacy, results from a meta-analysis of 4 studies failed to show significant benefit with disulfiram for preventing relapse or related outcomes (49).* Thus, other treatment options should be considered first and disulfiram should be reserved only for highly motivated patients and,

43. Amato L, Minozzi S, Davoli M. Efficacy and safety of pharmacological interventions for the treatment of the Alcohol Withdrawal Syndrome. Cochrane Database Syst Rev. 2011:CD008537. [PMID: 21678378] doi:10.1002/14651858.CD008537.pub2

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

ideally, when there is support to monitor medication adherence.

Acamprosate, FDA approved in 2004, is believed to work through the N-methyl-d-aspartate receptor to restore γ-aminobutyric acid and glutamate balance. The main challenges with this medication are that it is dosed 3 times daily and commonly leads to diarrhea and vomiting. In addition, as it is metabolized by the kidneys, dose adjustment is necessary in the setting of renal insufficiency. On the other hand, it is safe for use in patients with hepatic dysfunction in whom naltrexone may be contraindicated.

Results from a meta-analysis that pooled data from 16 studies found that to prevent 1 person from returning to any drinking, the number needed to treat (NNT) to achieve abstinence with acamprosate was 12 (CI, 8–26) (49). In contrast, based on data from 7 studies, acamprosate was not associated with decreasing the likelihood of return to heavy drinking.

Naltrexone was initially approved by the FDA as a once-daily oral medication in 1994 and then as a long-acting injectable medication in 2006. As an opioid-receptor antagonist, it works to decrease the reward pathways associated with alcohol use and decreases cravings. The main concerns with naltrexone are potential for worsening depression and suicidal ideation compared with placebo (www.accessdata.fda.gov/drugsatfda_docs/label/2010/021897s005s010lbl.pdf). Although patients and their families should be advised accordingly, trial data and clinical experience indicate that this is relatively uncommon. There is also concern for hepatotoxicity, particularly with the oral formulation that carries a warning for this risk, and periodic liver function test monitoring is appropriate, although such toxicity is rarely seen at the doses used to treat alcohol use disorder. In fact, on average, liver enzymes usually decrease in treated cohorts. Given the mechanism of action, opioids are contraindicated with naltrexone and patients need to be counseled before treatment initiation of this risk, and options for pain management should also be discussed if the need arises.

The NNT to prevent 1 person from returning to any drinking (n = 16 studies) and heavy drinking (n = 19 studies) with oral naltrexone was 20 (CI, 11–500) and 12 (CI, 8–26) respectively (49).

Injectable naltrexone is a good option for persons willing to receive a monthly injection because it can improve medication adherence. Naltrexone is also approved for the treatment of opioid use disorders and thus is a potential option for patients with concurrent alcohol and opioid use.

Studies investigating the efficacy of other medications, such as baclofen, gabapentin, ondansetron, topiramate, and varenicline, are under way with promising results (51).

Research consistently demonstrates that medications to treat alcohol use disorders and prevent relapse are underprescribed, despite their demonstrated effectiveness and availability.

For example, 1 study found that among 101 026 patients with an alcohol use disorder receiving care through the Veterans Health Administration, only 3% of patients received any form of naltrexone, 7% of whom received long-acting naltrexone (52).

These data, coupled with a mixed-methods evaluation of factors affecting implementation, support the need for increased provider awareness of the safety and effectiveness of these medications and skill in prescribing them (53). Ongoing training, such as that through the Substance Abuse and Mental Health Services Administration (www.samhsa.gov/medication-assisted-treatment and http://store.samhsa.gov/shin/content

44. Mirijello A, D'Angelo C, Ferrulli A, Vassallo G, Antonelli M, Caputo F, et al. Identification and management of alcohol withdrawal syndrome. Drugs. 2015;75:353-65. [PMID: 25666543] doi:10.1007/s40265-015-0358-1
45. Matching Alcoholism Treatments to Client Heterogeneity: Project MATCH posttreatment drinking outcomes. J Stud Alcohol. 1997;58:7-29. [PMID: 8979210]
46. Martin GW, Rehm J. The effectiveness of psychosocial modalities in the treatment of alcohol problems in adults: a review of the evidence. Can J Psychiatry. 2012; 57:350-8. [PMID: 22682572]
47. Humphreys K, Blodgett JC, Wagner TH. Estimating the efficacy of Alcoholics Anonymous without self-selection bias: an instrumental variables re-analysis of randomized clinical trials. Alcohol Clin Exp Res. 2014;38: 2688-94. [PMID: 25421504] doi:10.1111/acer.12557
48. Friedmann PD, Saitz R, Samet JH. Management of adults recovering from alcohol or other drug problems: relapse prevention in primary care. JAMA. 1998;279:1227-31. [PMID: 9555766]
49. Jonas DE, Amick HR, Feltner C, Bobashev G, Thomas K, Wines R, et al. Pharmacotherapy for adults with alcohol use disorders in outpatient settings: a systematic review and meta-analysis. JAMA. 2014; 311:1889-900. [PMID: 24825644] doi:10.1001/jama.2014.3628
50. Anton RF, O'Malley SS, Ciraulo DA, Cisler RA, Couper D, Donovan DM, et al; COMBINE Study Research Group. Combined pharmacotherapies and behavioral interventions for alcohol dependence: the COMBINE study: a randomized controlled trial. JAMA. 2006;295:2003-17. [PMID: 16670409]
51. Soyka M, Lieb M. Recent Developments in Pharmacotherapy of Alcoholism. Pharmacopsychiatry. 2015;48:123-35. [PMID: 25761458] doi:10.1055/s-0035-1547237

 © 2016 American College of Physicians

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

/SMA15-4907/SMA15-4907 .pdf), is useful for practicing clinicians seeking to improve their knowledge and confidence in prescribing these medications.

When should clinicians consider antidepressants or anxiolytics in the treatment of patients with unhealthy alcohol use?

Alcohol use disorders commonly co-occur with other substance use disorders and mental illness, including mood disorders, anxiety disorders, and personality disorders (2). Antidepressants are not effective stand-alone treatments for alcohol use disorders. These medications can, however, be effective for depressive symptoms among patients with unhealthy alcohol use, especially those with at-risk drinking (10, 54). Benzodiazepines are considered standard treatment for managing the acute phase of alcohol withdrawal but are not effective for alcohol use disorder and increase risk for an additional substance use disorder. Given mixed results regarding how to most effectively prioritize treatment, the alcohol use disorder and the co-occurring mental illness should be treated simultaneously with close monitoring for medication side effects and utilization of multimodal approaches (e.g., counseling) when possible (55).

Are there medications of particular concern in the setting of unhealthy alcohol use?

The **Box** (Examples of Common Alcohol–Prescription Medication Interactions) lists medications with potential interactions with alcohol. A more detailed list of over-the-counter and prescription medications that may interact with alcohol is available at http://pubs.niaaa.nih.gov /publications/Medicine/medicine .htm. Clinicians should note that these include medications commonly prescribed to treat lipid disorders (e.g., statins) and insomnia as well as nonsteroidal anti-inflammatory drugs, acetaminophen even at recommended doses, and anticoagulants.

Prescription opioids and benzodiazepines, both taken as prescribed or for recreation, are of great concern when mixed with alcohol. They should be prescribed judiciously (56), and patients should be counseled accordingly.

Alcohol contributes to approximately one fifth of prescription opioid–related and one fourth of benzodiazepine-related emergency department visits, respectively (57); it is involved in one fifth of deaths related to prescription opioids or benzodiazepines (57).

In addition, given the potential risk for harms associated with polypharmacy (i.e., receipt of 5 or more long-term medications), patients with unhealthy alcohol use and polypharmacy must be particularly closely monitored and carefully assessed to avoid potential harms (e.g., falls and medication nonadherence).

What additional care should be considered for patients with unhealthy alcohol use to promote health?

Patients with current alcohol use disorder have an increased prevalence of comorbid conditions that warrant screening and treatment. These include other substance use and mental health disorders, including tobacco use, drug use, anxiety, mood, and personality disorders (2). Prevalence of insomnia, anemia, osteoporosis, and liver disease is also increased. In addition to standard vaccinations as recommended for healthy adults (e.g., influenza), vaccination against hepatitis B and hepatitis C virus should be considered among those with established liver dis-

52. Marienfeld C, Iheanacho T, Issa M, Rosenheck RA. Long-acting injectable depot naltrexone use in the Veterans' Health Administration: a national study. Addict Behav. 2014;39:434-8. [PMID: 23790742] doi:10 .1016/j.addbeh.2013.05 .006
53. Harris AH, Ellerbe L, Reeder RN, Bowe T, Gordon AJ, Hagedorn H, et al. Pharmacotherapy for alcohol dependence: perceived treatment barriers and action strategies among Veterans Health Administration service providers. Psychol Serv. 2013;10:410-9. [PMID: 23356858] doi:10 .1037/a0030949
54. Nunes EV, Levin FR. Treatment of depression in patients with alcohol or other drug dependence: a meta-analysis. JAMA. 2004;291:1887-96. [PMID: 15100209]
55. Ipser JC, Wilson D, Akindipe TO, Sager C, Stein DJ. Pharmacotherapy for anxiety and comorbid alcohol use disorders. Cochrane Database Syst Rev. 2015;1:CD007505. [PMID: 25601826] doi:10 .1002/14651858.CD007505 .pub2
56. Chou R, Fanciullo GJ, Fine PG, Adler JA, Ballantyne JC, Davies P, et al; American Pain Society-American Academy of Pain Medicine Opioids Guidelines Panel. Clinical guidelines for the use of chronic opioid therapy in chronic noncancer pain. J Pain. 2009;10:113-30. [PMID: 19187889] doi:10 .1016/j.jpain.2008.10 .008
57. Jones CM, Paulozzi LJ, Mack KA; Centers for Disease Control and Prevention (CDC). Alcohol involvement in opioid pain reliever and benzodiazepine drug abuse-related emergency department visits and drug-related deaths - United States, 2010. MMWR Morb Mortal Wkly Rep. 2014;63: 881-5. [PMID: 25299603]

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

ease (www.cdc.gov/vaccines/schedules/downloads/adult/adult-pocket-size.pdf). Pneumococcal polysaccharide and zoster vaccination are recommended for those with an alcohol use disorder. Patients with longstanding alcohol use disorder can also experience cognitive and neurologic deficits, including peripheral neuropathy, Wernicke encephalopathy, and Korsakoff syndrome. Due to the increased risk for sexually transmitted infections associated with unhealthy alcohol use—including HIV (12)—patients should be appropriately screened, counseled, and offered prevention and treatment as indicated (58).

What type of follow-up care should clinicians provide for patients with unhealthy alcohol use, and when is specialty referral appropriate?

Patients with unhealthy alcohol use should be seen regularly for ongoing monitoring of alcohol use and treatment effects, as well as for any associated medical, psychiatric, and behavioral adverse effects. Referral to specialty care is appropriate for patients with evidence of at-risk alcohol use who do not respond to brief interventions, as well as those with an alcohol use disorder and/or significant comorbid medical or psychiatric conditions who have not responded to office-based treatments (4). The American Society of Addiction Medicine criteria can be helpful for guiding appropriate placement of patients who need a higher level of care (www.asam.org/publications/the-asam-criteria/about/).

58. US Public Health Service. Preexposure Prophylaxis for the Prevention of HIV Infection in the United States; 2014.

Treatment... For patients with at-risk drinking, brief interventions can be effective for enhancing motivation and decreasing alcohol use. For patients with an alcohol use disorder, treatment hinges on ensuring patient safety and stabilization. Benzodiazepines are the mainstay of treatment for decreasing alcohol withdrawal symptoms and risk of seizures; hospitalization is indicated for patients with moderate to severe withdrawal and high risk for complications. To prevent relapse, psychotherapeutic interventions (i.e., cognitive behavioral therapy, motivational enhancement therapy, or 12-step facilitation) in conjunction with pharmacotherapy (naltrexone or acamprosate) and self-help groups is recommended. Comprehensive care for all patients with unhealthy alcohol use should include optimizing medication regimens. Referral to specialty services, including addiction specialists, should be considered for patients who do not respond to treatment and for those with evidence of an alcohol use disorder or who have significant comorbidity.

CLINICAL BOTTOM LINE

Practice Improvement

What factors do U.S. stakeholders use to evaluate the quality of care for patients with unhealthy alcohol use?

The 2014 Centers for Medicare & Medicaid Services (CMS) Electronic Health Records Incentive Program includes the National Committee for Quality Assurance measure for initiation and engagement of alcohol and other drug dependence treatment. It indicates that the percentage of patients who initiated treatment within 14 days of a diagnosis of alcohol or drug dependence as well as the percentage of patients who initiated treatment

 © 2016 American College of Physicians

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

who had 2 or more additional services with and an alcohol or other drug dependence diagnosis within 30 days of the initial visit should be measured. The Health Resources and Services Administration HIV/AIDS Bureau contains a similar metric and indicates that all new patients with an HIV diagnosis be screened for alcohol use. Similarly, there is a quality measure supported by CMS, the National Quality Forum and the American Medical Association–Physician Consortium for Performance Improvement, which indicates that patients with hepatitis C virus should be counseled regarding alcohol use. These recommendations are consistent with the USPSTF guideline that adults aged 18 years or older be screened for unhealthy alcohol use and that individuals identified to have hazardous or at-risk drinking should be provided with brief behavioral counseling to reduce hazardous or at-risk drinking (20). Given the evidence (35), this is a grade B recommendation. The Veterans Health Administration's comprehensive clinical guidelines state that all patients seen in general medical and mental health settings should be screened for unhealthy alcohol use (59). Those who meet criteria for unhealthy alcohol use should receive a brief counseling intervention with consideration of referral to specialty addiction programs at the initial visit or follow-up. These guidelines also provide algorithms for treatment, including the role of pharmacotherapy.

What clinical practice changes are occurring as a result of the Affordable Care Act?

With the passing of the Affordable Care Act, treatment of alcohol use disorders is considered an essential health benefit and insurance coverage for such services is required. As such, there have been important initiatives to improve access to and delivery of treatment of unhealthy alcohol use and other substance use through routine clinical settings. In addition, efforts to determine how best to measure and improve the quality of screening and treatment delivery for unhealthy alcohol use are actively under way (60). Given the significant individual and public health impact of unhealthy alcohol use, such initiatives are welcomed and warranted.

59. Veterans Health Administration/Department of Defense. VHA/DoD Clinical Practice Guideline for the Management of Substance Use Disorders in the Primary Care Setting; 2009.
60. Harris AH. The primitive state of quality measures in addiction treatment and their application [Editorial]. Addiction. 2015. [PMID: 26395364] doi:10.1111/add.13096

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

In the Clinic

Tool Kit

Alcohol Use

Patient Information

http://pubs.niaaa.nih.gov/publications/Practitioner/CliniciansGuide2005/guide.pdf
www.niaaa.nih.gov/publications/brochures-and-fact-sheets
National Institute on Alcohol Abuse and Alcoholism
www.cdc.gov/alcohol/fact-sheets.htm
Centers for Disease Control and Prevention

Clinical Guidelines

http://annals.org/article.aspx?articleid=1722524&resultClick=3
U.S. Preventive Services Task Force
www.guideline.gov/content.aspx?id=23784
Agency for Health Care Research and Quality
https://www.nice.org.uk/guidance/cg115
National Institute for Health and Care Excellence

Community-Based Resources for Treatment of Alcohol Misuse

www.aa.org/
Alcoholics Anonymous with information available in English, Spanish, and French
www.samhsa.gov/atod/alcohol
National Clearinghouse for Alcohol and Drug Information

In the Clinic

 © 2016 American College of Physicians

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

WHAT YOU SHOULD KNOW ABOUT ALCOHOL USE

How Can Alcohol Affect Me?

According to the NIAAA, for men up to age 65 years lower-risk alcohol use is no more than 4 drinks on any single day and no more than 14 drinks per week. For women, who tend to have lower volumes of distribution, and men older than 65 years, lower-risk alcohol use is no more than 3 drinks on any single day and no more than 7 drinks per week

When Is Alcohol Use Unhealthy?

Unhealthy alcohol use can cause serious health and emotional problems. Drinking more than what is outlined above is considered unhealthy alcohol use. Unhealthy alcohol use can cause:

- Reflux
- Ulcers
- Vitamin deficiencies
- High blood pressure
- Various cancers
- Stroke
- Cirrhosis (scarring of the liver)
- Heart Problems
- Seizures
- Alcohol poisoning
- Depression
- Diabetes mellitus

Unhealthy alcohol use is also associated with increased risk for diabetes, HIV and other sexually transmitted infections, and depression. It can lead to unhealthy behaviors that may cause:

- Car accidents
- Accidents like falling or drowning
- Firearm injuries
- Intimate partner violence
- Homicide
- Suicide

How Is Unhealthy Alcohol Use Diagnosed?

Your doctor will ask you about your medical history and complete a physical examination. He or she may take blood samples, which will help the doctor know if your drinking is causing health problems.

How Is Unhealthy Alcohol Use Treated?

Your doctor or other health professional can support you in cutting down on drinking. Family and friends can help, too. There are other ways to reduce alcohol use, including:

- Talk therapy
- 12-step programs or Alcoholics Anonymous
- Certain medicines

How Can I Use Alcohol Safely?

Talk with your doctor about whether it is safe for you to drink alcohol if you:

- Take prescription medicines
- Are older than age 65
- Are a young adult
- Have a chronic health condition

For More Information

ACP[SM]
American College of Physicians
Leading Internal Medicine, Improving Lives

Centers for Disease Control and Prevention
www.cdc.gov/alcohol/fact-sheets/alcohol-use.htm

National Institute on Alcohol Abuse and Alcoholism
www.niaaa.nih.gov/alcohol-health/overview-alcohol-consumption

Patient Information

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016

Appendix Figure. **Recommended screening instruments: The NIAAA Single-Question Screening Item, AUDIT, and AUDIT-C questionnaires.***

NIAAA Single-Question Screening Item

Question: How many times in the past year have you had *x* or more drinks in a day? (where *x* is 5 for men, 4 for women and 1 standard drink is equivalent to 12 ounces of beer, 5 ounces of wine, or 1.5 ounces of 80-proof spirits)

Scoring: One or more episodes is considered a positive screen.

AUDIT

Questions	Points 0	1	2	3	4
1. How often do you have a drink containing alcohol?	Never	Monthly or less	2–4 times a month	2–3 times a week	4 or more times a week
2. How many drinks containing alcohol do you have on a typical day when you are drinking?	1–2	3–4	5–6	7–9	10 or more
3. How often do you have 5 or more drinks on one occasion?	Never	Less than monthly	Monthly	Weekly	Daily or almost daily
4. How often during the last year have you found that you were not able to stop drinking once you started?	Never	Less than monthly	Monthly	Weekly	Daily or almost daily
5. How often during the last year have you failed to do what was normally expected of you because of drinking?	Never	Less than monthly	Monthly	Weekly	Daily or almost daily
6. How often during the last year have you needed a first drink in the morning to get yourself going after a heavy drinking session?	Never	Less than monthly	Monthly	Weekly	Daily or almost daily
7. How often during the last year have you had a feeling of guilt or remorse after drinking?	Never	Less than monthly	Monthly	Weekly	Daily or almost daily
8. How often during the last year have you been unable to remember what happened the night before because of your drinking?	Never	Less than monthly	Monthly	Weekly	Daily or almost daily
9. Have you or someone else been injured because of your drinking?	No		Yes, but not in the last year		Yes, during the last year
10. Has a relative, friend, doctor, or other health care worker been concerned about your drinking or suggested you cut down?	No		Yes, but not in the last year		Yes, during the last year

Scoring: Points for each of the ten items are added together. Total score of ≥8 for men up to age 60 years or ≥4 for women, adolescents, and persons older than 60 years are considered positive. Cutoffs may vary depending on purpose and population of interest.

AUDIT-C

Questions	Points 0	1	2	3	4
1. How often did you have a drink containing alcohol in the past year?	Never	Monthly or less	2–4 times a month	2–3 times a week	4 or more times a week
2. How many drinks did you have on a typical day when you were drinking in the past year?	None or 1–2	3–4	5–6	7–9	10 or more
3. How often did you have 5 or more drinks on one occasion in the past year?	Never	Less than monthly	Monthly	Weekly	Daily or almost daily

Scoring: Points for each of the three items are added together. Total score of ≥6 for men and ≥4 for women is considered positive. Cutoffs may vary depending on purpose and population of interest.

AUDIT = Alcohol Use Disorders Identification Test; NIAAA = National Institute on Alcohol Abuse and Alcoholism.

* Figure adapted from reference 5.

© 2016 American College of Physicians

Downloaded From: http://annals.org/ by a University of California San Diego User on 12/09/2016